

Bypassing the Haar Singularity: Natural Density Uniformity of the Collatz Discrepancy Function

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Abstract

The Haar measure on $\mathbb{Z}_2$, although preserved by the Syracuse map, is singular with respect to the counting measure on $\mathbb{N}$. This prohibits direct transfer of 2-adic almost-everywhere results to integer orbit properties. We introduce M7, which circumvents this obstruction by working entirely in the integer counting measure (natural density). We define the discrepancy function $\Delta(O) = R - R_{\min}(O)$ and conjecture that $\Delta(O) \bmod p$ becomes uniformly distributed as $O \rightarrow \infty$. This provides a probabilistically valid channel for finite-field sieving that is independent of Haar measure, opening the door for probabilistic number theory methods in Collatz cycle exclusion.

1. The Haar Obstruction

A beautiful and well-established result in Collatz theory (Kontorovich-Lagarias 2009, M1) states that the Syracuse map T preserves the Haar measure μ_2 on $\mathbb{Z}_2$. This has led to powerful "almost everywhere" results: Tao (2019) proves boundedness for μ_2 -almost all 2-adic integers, and Terras (1976) proves eventual descent for μ_2 -almost all integers.

However, as established in M2 and M4, μ_2 is singular with respect to the counting measure on $\mathbb{N}$ ($\#_N$). The two measures are mutually singular. Consequently:

- μ_2 -almost-everywhere does not imply natural-density-almost-everywhere in $\mathbb{N}$.
- 2-adic spectral properties cannot be directly pulled back to integer frequency spectra.
- This is the fundamental reason why distributional results do not imply pointwise convergence.

2. The Natural Density Framework

Let $A \subset \mathbb{N}$. The natural density (or asymptotic density) is defined as the limit as N approaches infinity of $|A \cap \{1, \dots, N\}| / N$, if the limit exists. This is the appropriate measure for integer-orbit statements. Our goal is to formulate statements about Collatz dynamics directly in this measure, without reference to $\mathbb{Z}_2$ or Haar measure.

3. The Discrepancy Function

Recall the cycle accumulator decomposition:

$$R = R_{\min}(O) + \Delta(O)$$

where $R_{\min}(O) = (6^O - 1)/5$ and $\Delta(O) = \sum_{j=0}^{O-1} 3^j (2^{e_j} - 2^j)$.

The distribution of $\Delta(O) \bmod p$ is determined by the distribution of $e_j \bmod \text{ord}_p(2)$. From P2, the e_j are geometrically distributed in the Haar sense. The key question: does this extend to natural density?

4. The Uniformity Conjecture (M7)

Conjecture (M7, Δ -uniformity): For any prime p , the sequence $\Delta(O) \bmod p$ is uniformly distributed on F_p under the natural counting density.

Supporting heuristic: Since P2 implies that $v_2(3n+1)$ is geometrically distributed, we expect that $e_j \bmod \text{ord}_p(2)$ behaves like independent random variables with uniform distribution on the cyclic group. Then $\Delta(O) \bmod p$ is a sum of O independent, identically distributed random variables, and the law of large numbers on finite groups gives uniformity as $O \rightarrow \infty$.

Technical requirement: Rigorous proof requires non-trivial estimates of 2-adic exponential sums which may be approached via 2-adic harmonic analysis or mixed character sum bounds.

5. Application to Cycle Exclusion

If M7 holds, then for any fixed prime p and any target residue c , the set of O for which $\Delta(O) \equiv c - R_{\min}(O) \pmod{p}$ has natural density $1/p$. In particular, for a large ensemble of primes, the density of O satisfying the full sieve condition decays as the product of $1/p_j$.

Combining with P34, the number of (O, S) pairs passing the modular sieve for m primes grows sub-quadratically, and the natural density of surviving cycles tends to 0 as m approaches infinity.

6. Comparison with Haar-Based Approaches

- **Haar measure on $\mathbb{Z}_2$:** Valid for 2-adic integers (almost-everywhere in μ_2). Limitation: Cannot pull back to $\mathbb{N}$.
- **Natural density (M7):** Valid for Integers $\mathbb{N}$ (almost-everywhere in counting measure). Limitation: Requires uniformity proof.
- **Direct pointwise:** Valid for each $n \in \mathbb{N}$ (for all n). Limitation: Unproven (Collatz itself).

M7 occupies a middle ground: it is stronger than Haar results for integer applications, but weaker than full pointwise proof.

7. Conclusion

M7 offers a rigorous path to bypass the Haar-counting singularity. The conjecture that $\Delta(O) \bmod p$ is uniformly distributed is plausible and provides a probabilistic sieve for cycle exclusion. Its proof would open the door for probabilistic number theory methods to contribute to Collatz cycle analysis without invoking the non-transferable Haar measure.

References

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